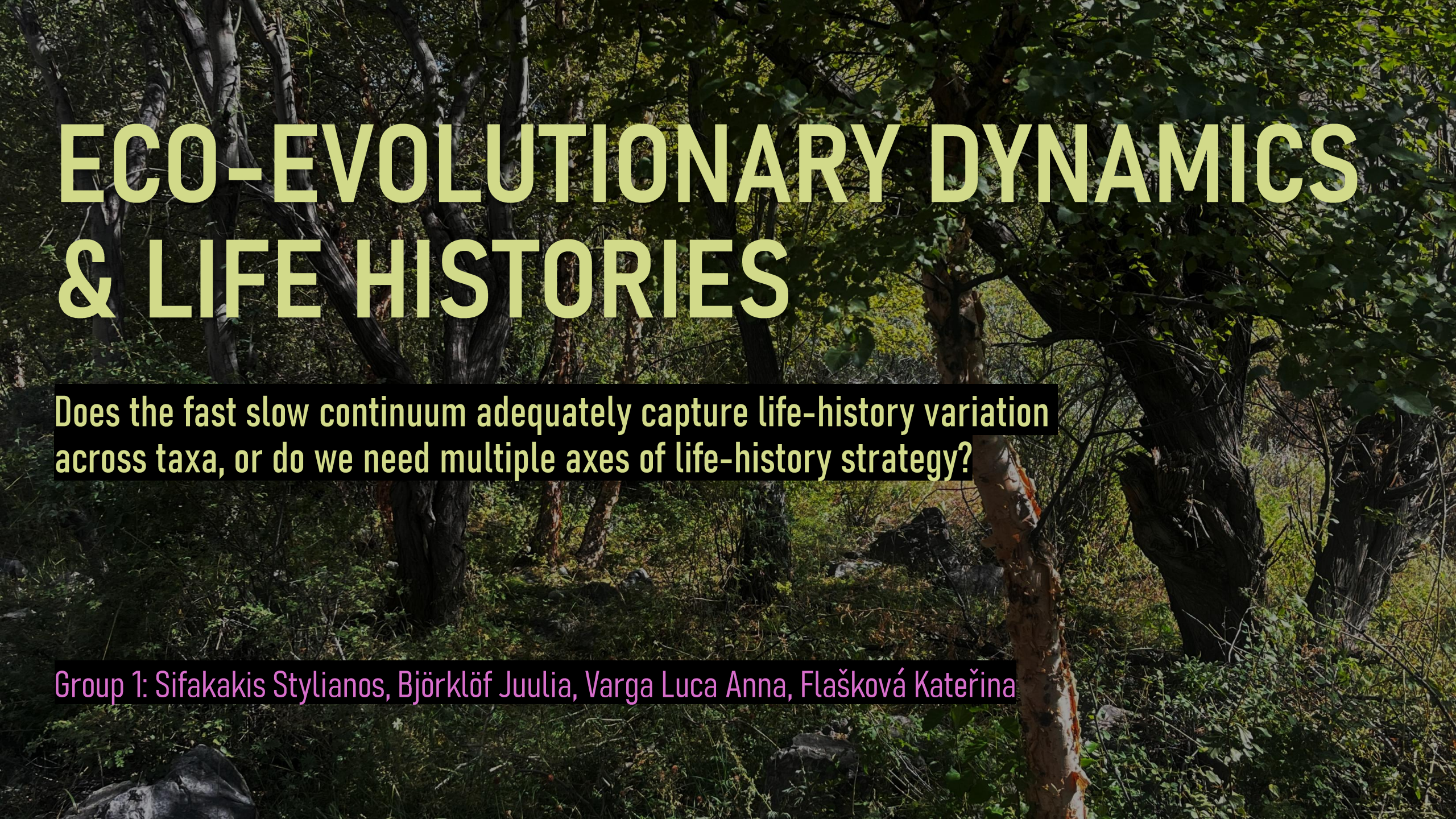
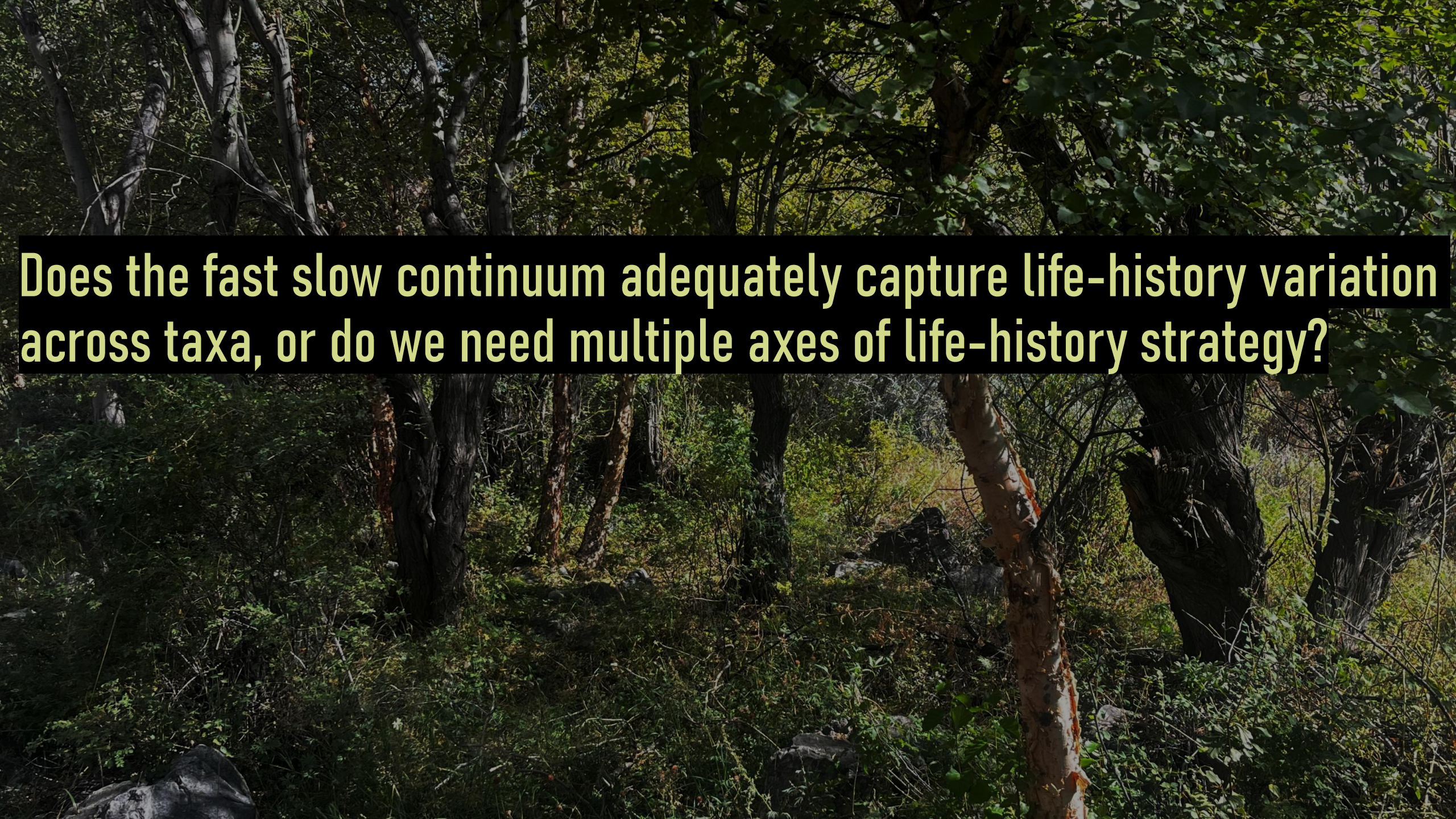


# ECO-EVOLUTIONARY DYNAMICS & LIFE HISTORIES

Does the fast slow continuum adequately capture life-history variation across taxa, or do we need multiple axes of life-history strategy?

Group 1: Sifakakis Stylianos, Björklöf Juulia, Varga Luca Anna, Flašková Kateřina


Does the fast slow continuum adequately capture life-history variation across taxa, or do we need multiple axes of life-history strategy?

# WHAT ARE ECO-EVOLUTIONARY DYNAMICS?



PHOTOGRAPH FROM BILDAGENTUR ONLINE/MCPHOTO-ROLFE/ALAMY

# LIFE HISTORY



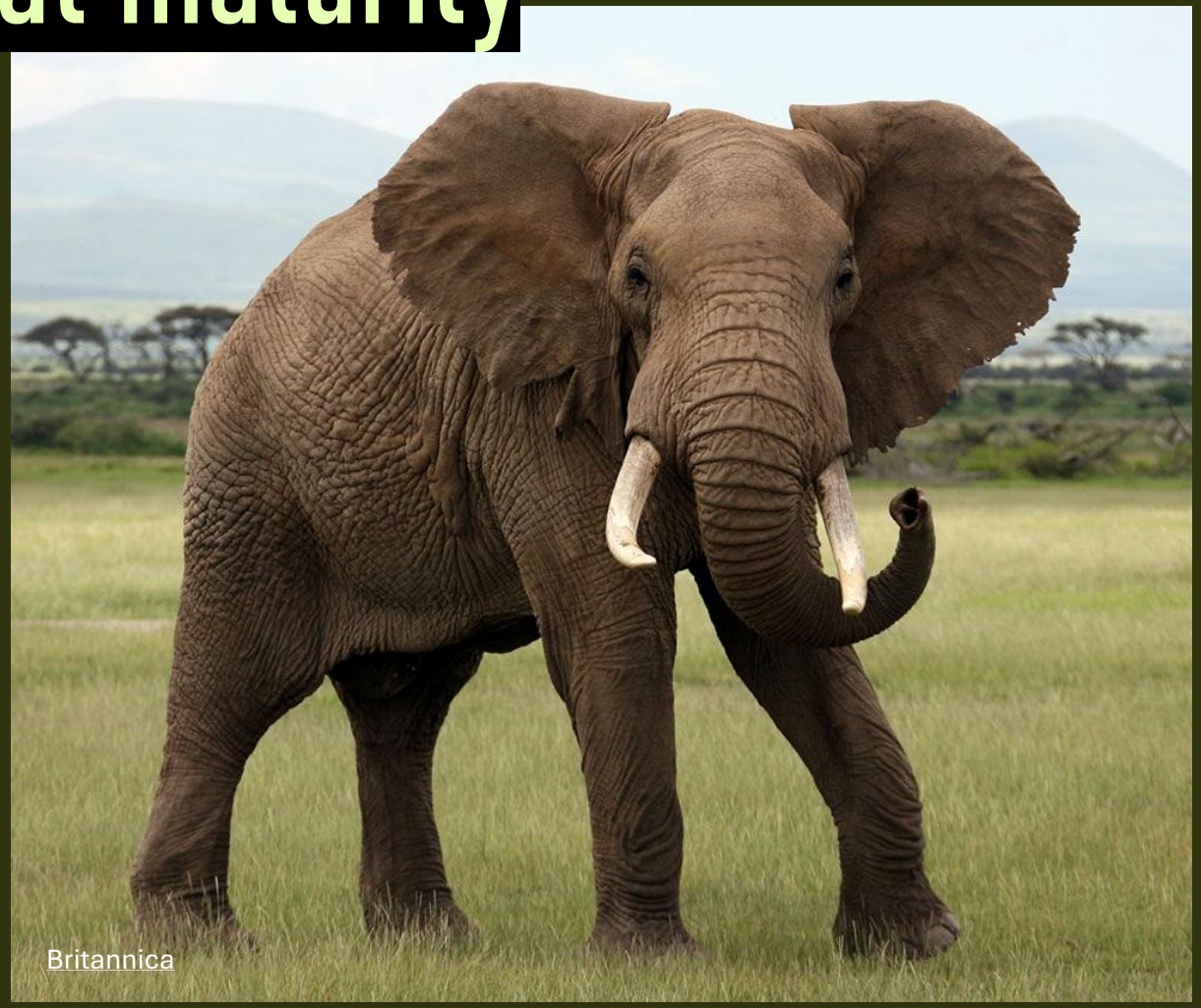
# LIFE HISTORY

## Life history traits

# Age and size at maturity




Eurasian harvest mouse
(Micromys minutus)




African bush elephant
(Loxodonta africana)

# Parity



Wikipedia

Sockeye salmon
(Oncorhynchus nerka)



euqirneto / Wikimedia Commons (CC BY 4.0)

# Parental investment



Britannica

Acacia tree on a savanna in Zimbabwe.



Via Mr. Canary Company (Odell, *Mom, Mom, MOMMM!*, 2020)

American robin
(Turdus migratorius)


LIFE HISTORY
Life history traits
Life history strategies

# Life history variation

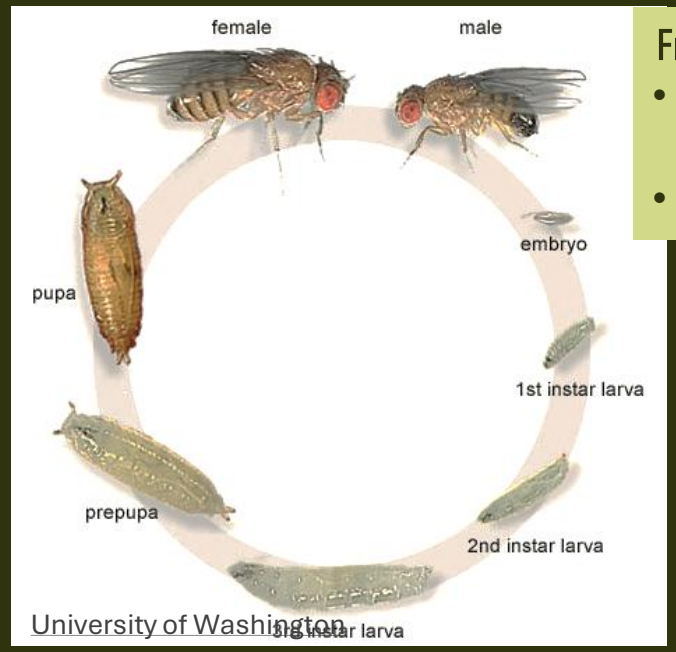
→ Limited combinations

→ axis defined by the trade-off between allocation of time to reproduction and development versus survival

# FAST-SLOW CONTINUUM

Two extremes:





University of Washington

Fruit fly (Drosophila melanogaster)

- Life cycle: 10-12 days in room temperature
- Genetic studies


kids.nationalgeographic.com

Przewalski horse

- Age at maturity: 4-5 years old
- Average lifespan: 20 years

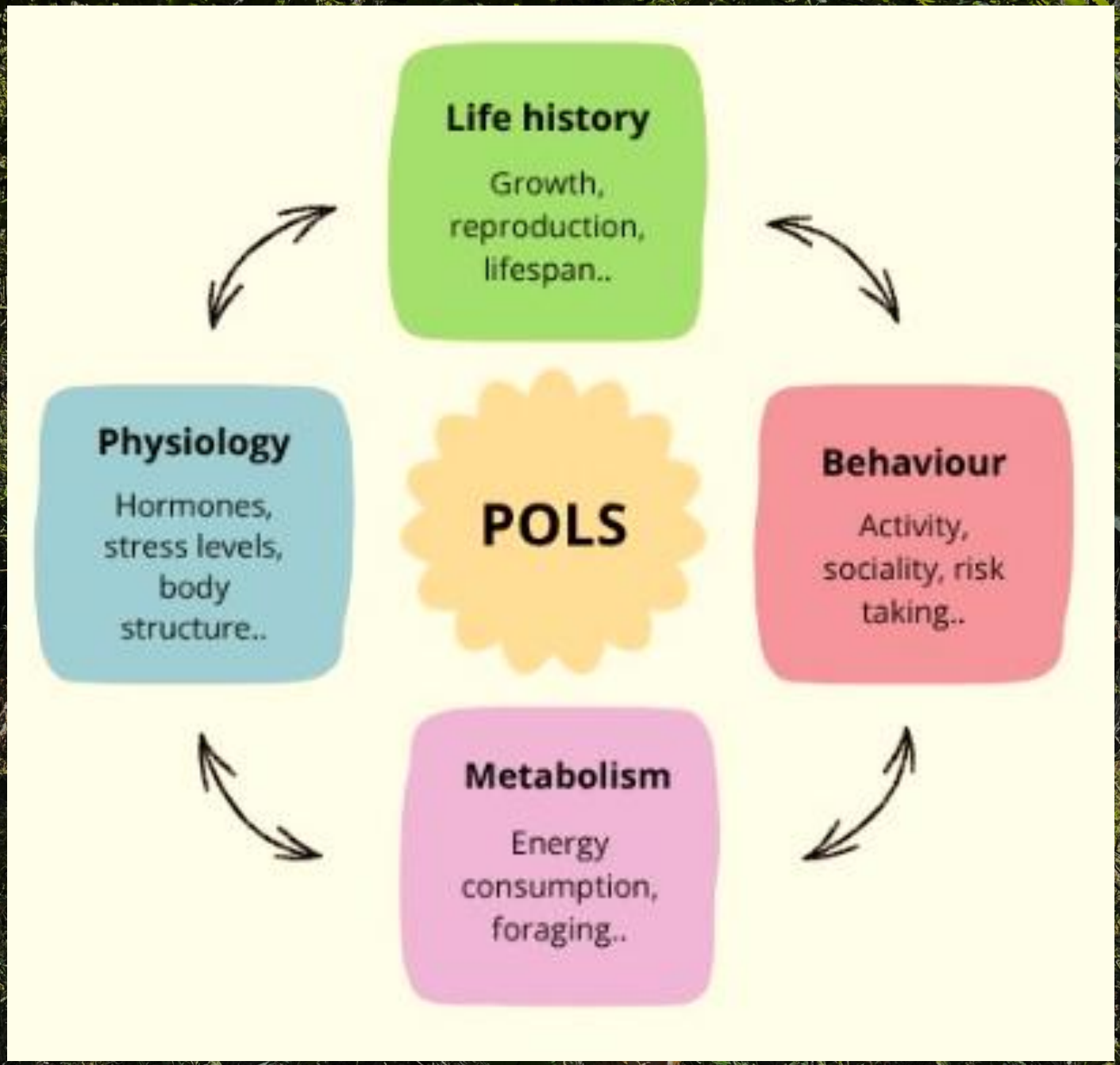
## FAST CONTINUUM

- Fast development
- Earlier maturation
- Earlier reproduction
- More offsprings
- Shorter lifespan

## SLOW CONTINUUM

- Slower development
- Later maturation
- Later reproduction
- Less offsprings
- Longer lifespan

# PACE-OF-LIFE SYNDROMES





Britannica



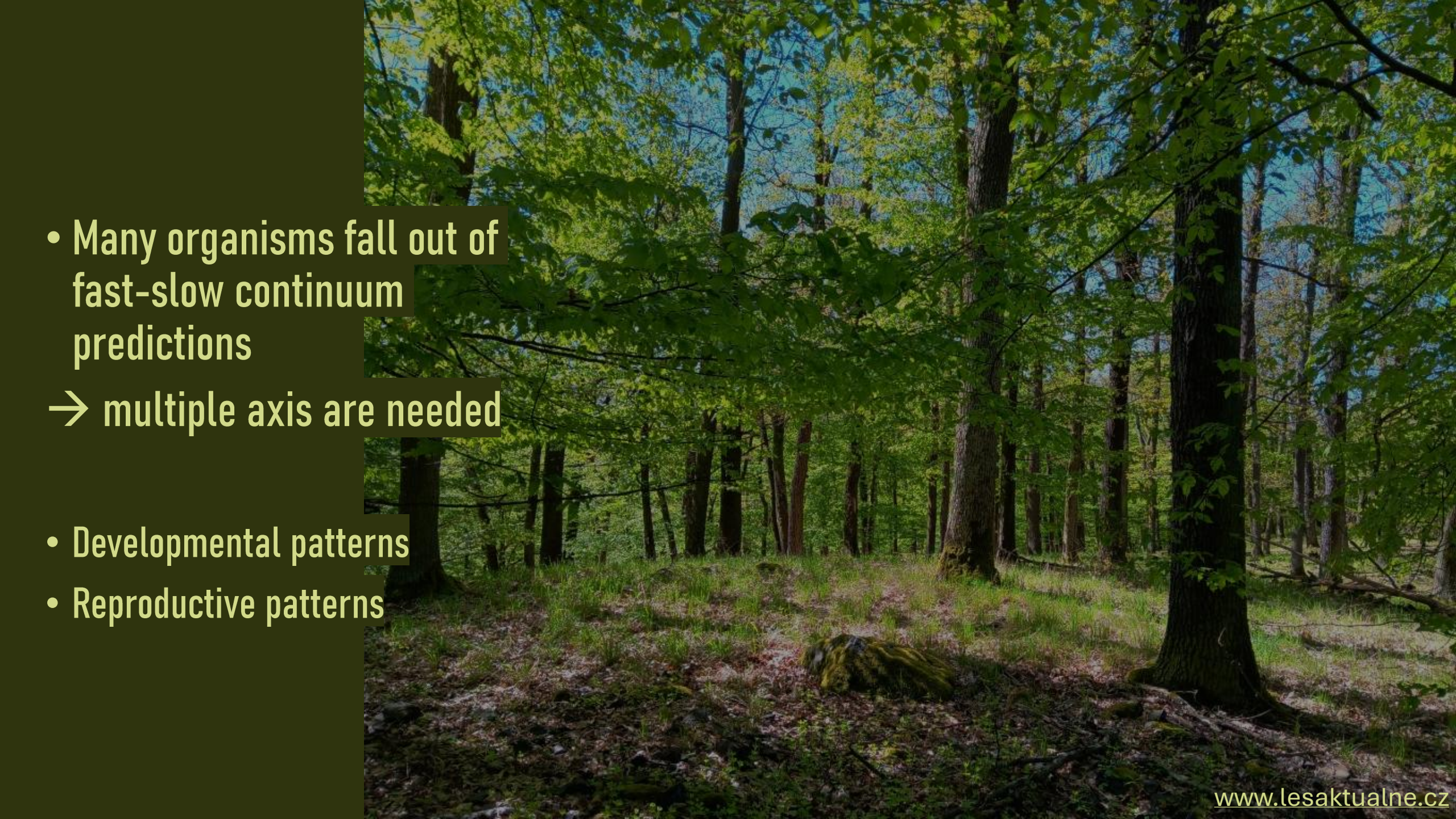
a-z-animals

# Is the fast-slow continuum enough to define variation of life history across taxa?

- Dominant axis

- Broad picture in tetrapods
- Vascular plants and corals

- Few studies in invertebres (except insects)



- Many organisms fall out of fast-slow continuum predictions

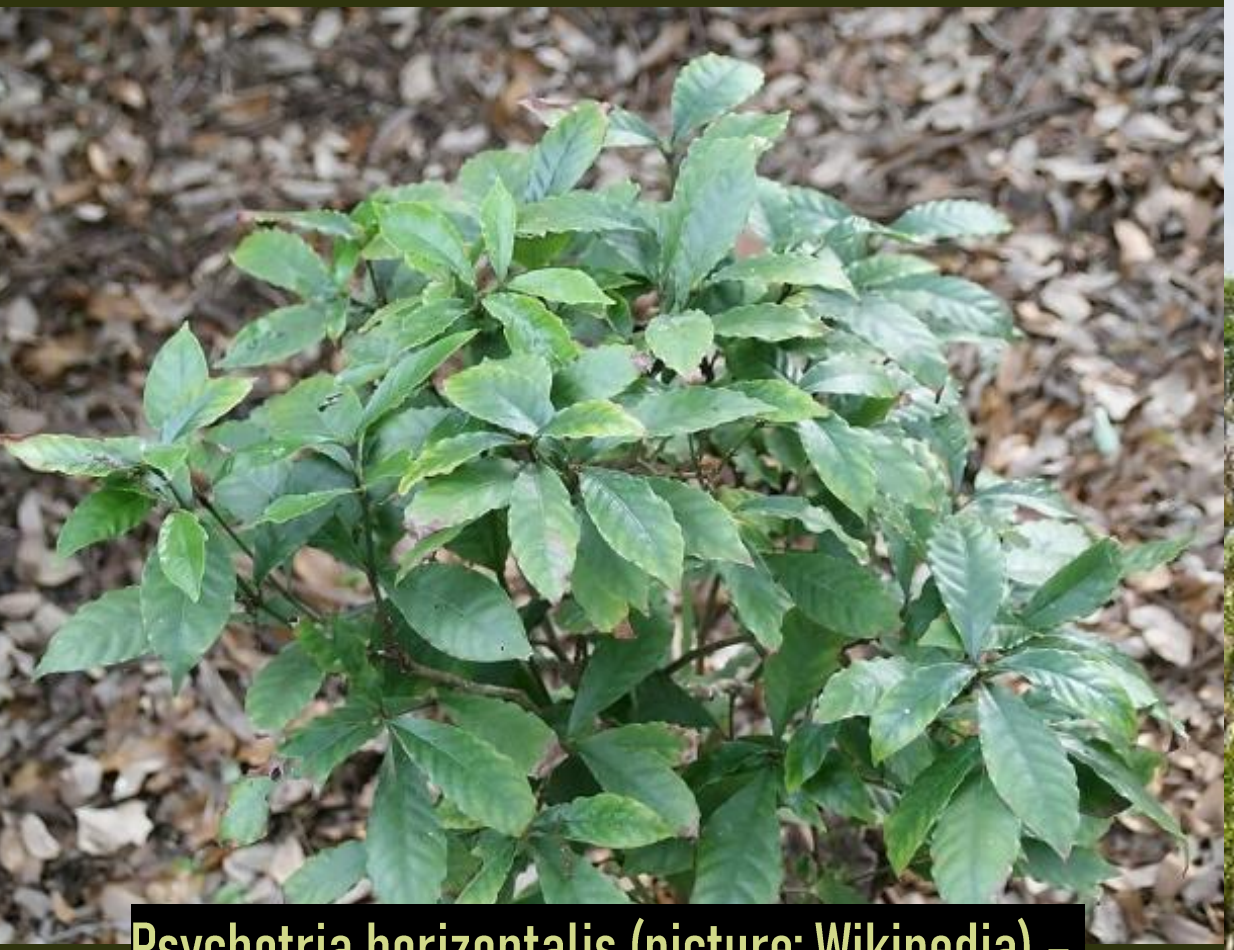
→ multiple axis are needed

- Developmental patterns
- Reproductive patterns

www.lesaktualne.cz

# High vs. low recruitment rates (stature–recruitment' axis)

→ rate of the addition of new individuals to a defined segment of a population



Psychotria horizontalis (picture: Wikipedia) – high recruitment rate (low juvenile mortality)



Ceiba pentandra (kapok tree) (picture: Ceiba foundation) – low recruitment rate (high juvenile mortality)

# Altricial offspring vs. Precocial offspring



PARROT

PELICAN

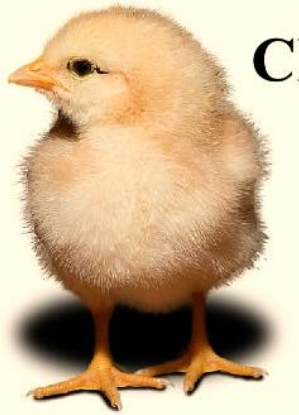




SONGBIRDS

Wikipedia

CHICKEN



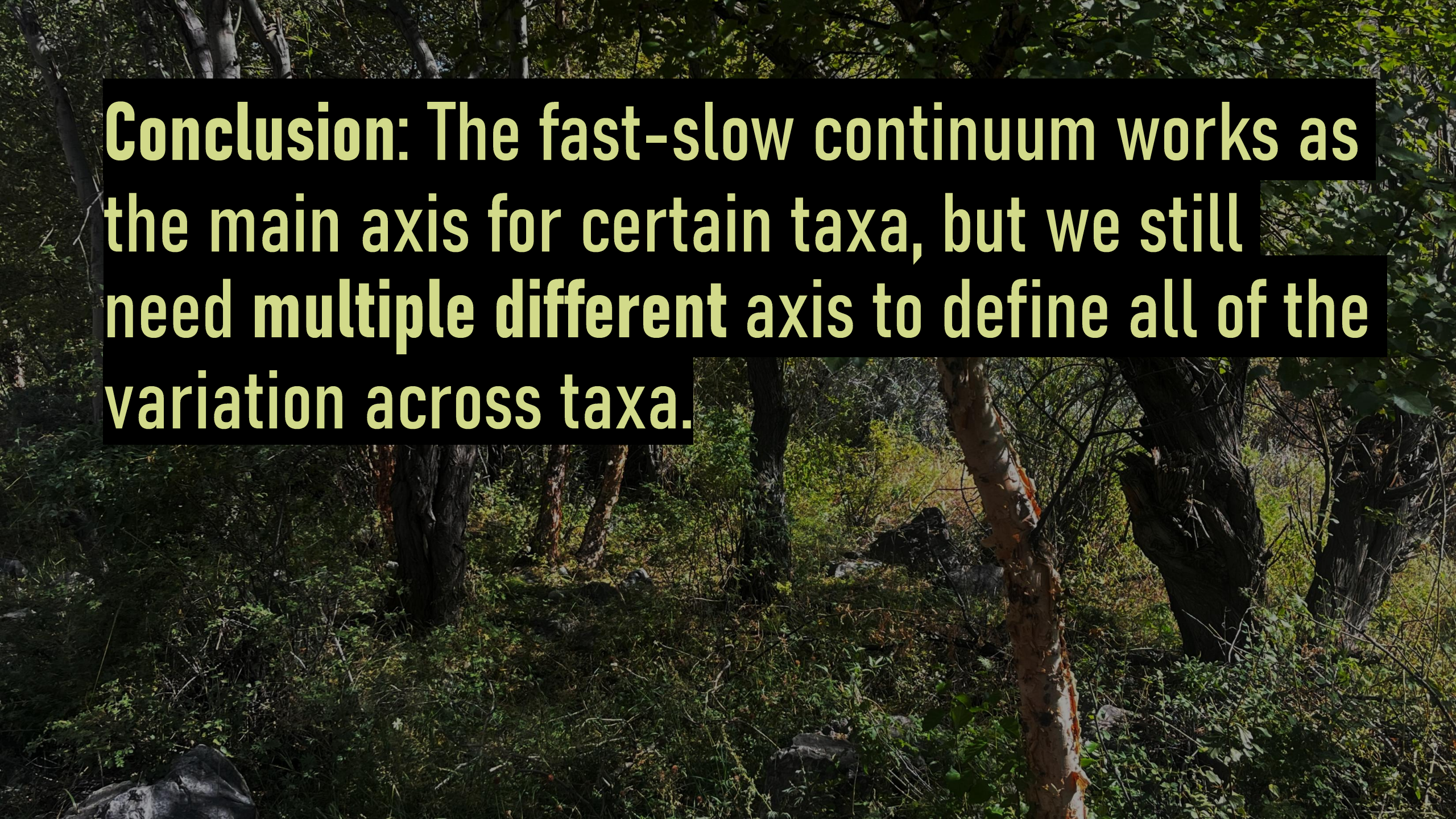
DUCK



RHEA



Wikipedia


**Conclusion**: The fast-slow continuum works as the main axis for certain taxa, but we still need **multiple different** axis to define all of the variation across taxa.


QUESTIONS?

# Resources


Salguero-Gómez, R. (2016). Article on plant life-history strategies. New Phytologist.

Les aktuálně. (n.d.). Jak duby snášejí klimatické extrémy.

National Geographic. (n.d.). Przewalski's Horse.

Wikipedia contributors. (n.d.). Krytosemenné. Wikipedie.

Pacific Salmon - Wild Salmon Center

Rabbit facts | Mammals | BBC Earth | BBC Earth

American Robin: Facts, Diet, Habitat, Behavior & Identification Guide

European rabbit — Mammal Society

Elephant Facts | Mammals | BBC Earth | BBC Earth

Eurasian Harvest Mouse - Facts, Diet, Habitat & Pictures on Animalia.bio

Arnot, M. (2017). A Response to Gardner and Harrison: The Reasons for, and Implications of, Brexit from an Anthropological Standpoint. ResearchGate.

ScienceDirect research article. (2020).

Royal Botanic Gardens, Kew. (n.d.). Plants of the World Online: Taxon record.

Nature Communications research article. (2024). Nature Communications.

# Resources

Stearns, S. C. (1977). The evolution of life history traits: a critique of the theory and a review of the data. Annual Review of Ecology and Systematics, 145–171.

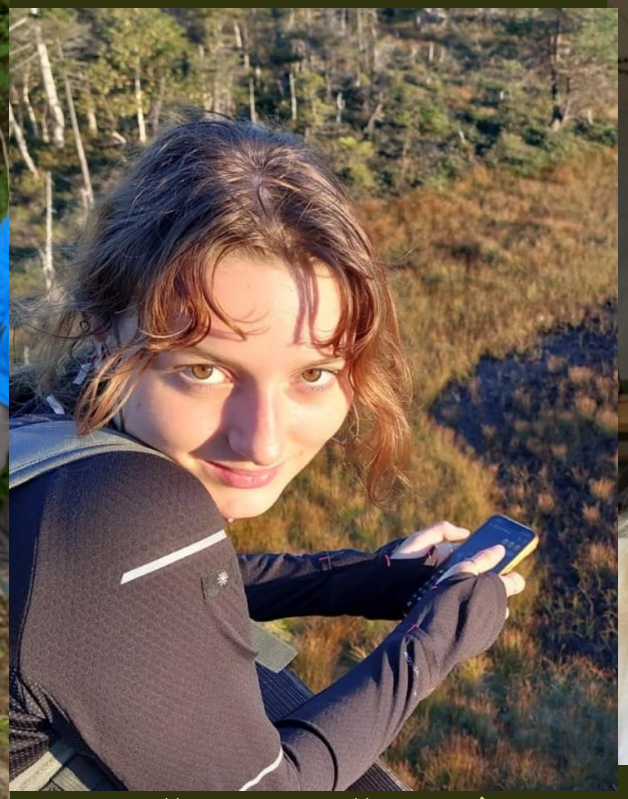
Stott, I. et al. (2024). Life histories are not just fast or slow. Trends in Ecology & Evolution, 39(9), 830–840.

# Contributions



## Varga Luca Anna

- Extract information
- Presentating
- Conseptualisation



## Kateřina Flašková

- Presentation and notes
- Extract information
- Presentating
- Conseptualisation



## Juulia Björklöf

- Conseptualisation
- Extract information
- Presentating




## Stylianos Sifakakis

- Presentation and notes
- Extract information
- Presentating
- Conseptualisation